Part 4 — Trusting the numbers: economics, method validation, omics as constraints, and getting DNA into a wild strain (weeks 49 to 52)

These four weeks are about whether your numbers mean anything. Week 49 asks whether the process clears its market price, which is the calculation that kills more strains than any reviewer. Week 50 asks whether the titer you reported is real. Week 51 asks what an omics dataset is actually evidence for, and how to turn it into a constraint rather than a figure. Week 52, which the tracker marks optional, asks why a wild industrial strain destroys your plasmid on arrival and what a pooled screen can and cannot tell you.

Like part 3, none of this has a textbook behind it. Unlike part 3, the reason is not that the subject is young: analytical validation and discounted cash flow are old and well codified, but the codification lives in regulatory guidance documents and process-economics texts that no one hands a biologist, and the translation into what a strain engineer actually needs has not been written down. So the chapters do the translation, with the arithmetic worked.

Two of these weeks will make you unpopular if you use them carelessly. A techno-economic analysis done honestly on a colleague's favourite strain usually says no, and a method validation done honestly on your own early data usually says the first six months of titers were not measurable. Both are worth doing anyway, and both are much cheaper done early. The way to make them useful rather than destructive is in chapter 49's last section: report what would have to be true for the answer to change, rather than only the answer.

Ch.	Week	Chapter
49	49	Techno-economic analysis and life-cycle assessment: the calculation that kills strains
50	50	Analytical method validation: trusting your own numbers
51	51	Omics, screening, and turning data into model constraints
52	52	Host defences and pooled screens: getting DNA into a wild strain, and reading 50,000 of them at once

Techno-economic analysis and life-cycle assessment: the calculation that kills strains

A techno-economic analysis is not a business exercise appended to a science project. It is the only calculation that tells you which of titer, rate and yield to work on next, and it takes an afternoon.

What a flowsheet model contains that week 18's envelope did not

Week 18 gave you the back-of-envelope: feedstock cost divided by yield, broth volume per kilogram, annual output per vessel. That is enough to rank strains and not enough to decide a process, because it omits four things that frequently dominate.

- **Capital, properly sized.** Not just the fermenters, but the seed train, the separation sequence, the utilities, and the installation that turns a $200,000 piece of equipment into a $700,000 installed item.
- **Utilities with a physical basis.** Agitation power, air compression, cooling duty — all of which follow from the oxygen demand you computed in chapter 19 rather than from a guess.
- **Waste.** Spent broth, biomass, salt from pH control, solvent losses from chapter 45. In a commodity process this is a cost line and occasionally a permit problem.
- **The cost of money.** Capital spent now against product sold over twenty years, which is a discounted cash flow and not a division.

The tool for this is BioSTEAM, which is free, Python, and comes with complete biorefinery models you can run unmodified before building your own. Run one of those first. The value of running someone else's model is seeing the line items you would have forgotten.

The model this chapter uses, stated in full

Everything below comes from one small model, and it is stated so you can disagree with it. Target output 10,000 t/yr. Fermenters of 200 m³ working volume, 48 h batch plus 8 h turnaround, 8,000 operating hours a year. Yield 0.40 g/g on glucose at $0.40/kg. Installed capital of $3M per fermenter train slot, plus $120 per m³ of annual broth throughput for the downstream train. Utilities $0.15/kg of product, labour $3M/yr, maintenance at 4% of capital, a 10% discount rate over a 20-year life. The capital recovery factor is then

```
CRF = i(1 + i)^n / ((1 + i)^n − 1) = 0.1175 at i = 0.10, n = 20
```

so each dollar of capital costs about 11.75 cents a year, and the capital charge per kilogram is CRF × capital ÷ annual output. Minimum product selling price is the sum of feedstock, utilities, fixed costs and that capital charge — the price at which net present value is zero, which is the number to compare against the market.

What it says: three titers, one conclusion

Titer	Fermenters	Broth per year	Capital	MPSP	Breakdown ($/kg)
20 g/L	17.5	500,000 m³	$112.5M	**$3.22/kg**	feed 1.00, util 0.15, fixed 0.75, capital 1.32
50 g/L	7.0	200,000 m³	$45.0M	**$2.16/kg**	feed 1.00, util 0.15, fixed 0.48, capital 0.53
100 g/L	3.5	100,000 m³	$22.5M	**$1.80/kg**	feed 1.00, util 0.15, fixed 0.39, capital 0.26

Against an incumbent price of $1.80/kg — a typical commodity number — the 20 g/L process loses money by 79%, the 50 g/L process by 20%, and the 100 g/L process breaks even exactly. That is the whole argument for the industrial obsession with titer, and notice what produces it: the feedstock term does not move at all, because yield is unchanged. Titer moves capital and fixed costs, by reducing the number of vessels and the volume of broth to process. Yield moves feedstock. They are different levers acting on different lines, which is why the question "should we work on titer or yield" has an arithmetic answer rather than a cultural one.

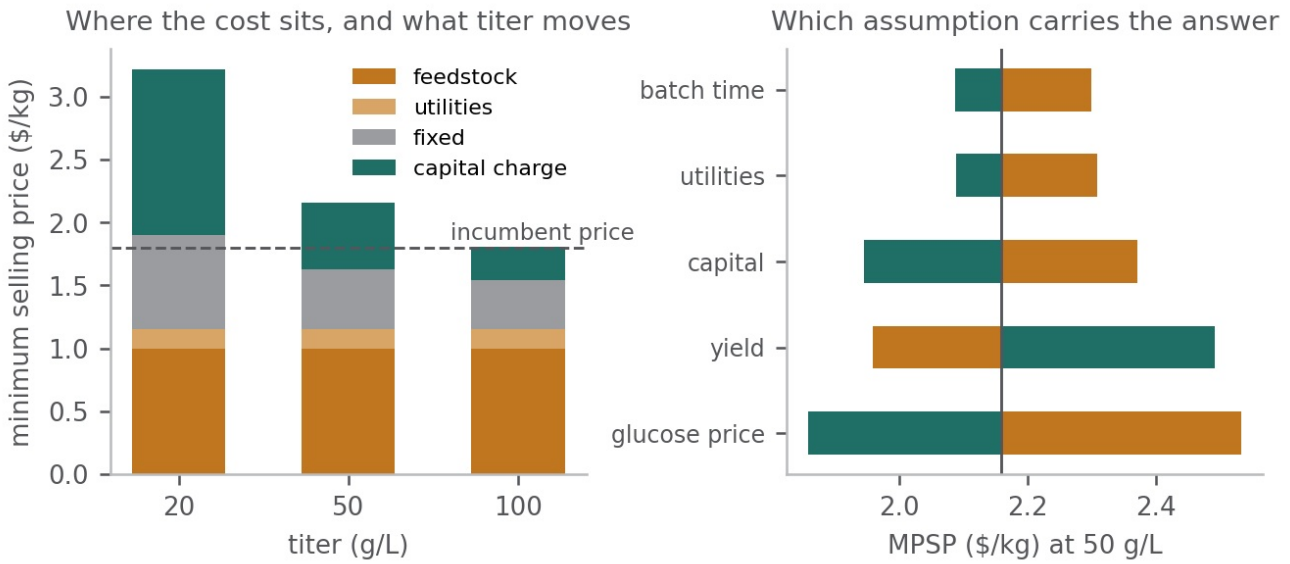


Figure 49.1 Left: the cost breakdown at three titers, against an assumed incumbent price. Right: the tornado plot at 50 g/L, each bar spanning the stated range of one input with everything else held at base. Computed from the model stated above; every parameter in it is an assumption, and the point of the right-hand panel is to show which assumptions matter.

Sensitivity: which assumption is carrying the conclusion

The tornado plot ranks the swings at 50 g/L: glucose price between $0.28 and $0.55/kg moves MPSP by $0.68/kg, yield between 0.30 and 0.50 g/g by $0.53, capital ±30% by $0.43, utilities by $0.22, and batch time between 36 and 72 h by $0.21. So the conclusion at 50 g/L is carried by a feedstock price you do not control and a yield you do. That is the sentence to put in the discussion, and it also tells you that arguing about batch time is arguing about the fifth-largest term.

Beyond ranking, there is the question of what to report. A single MPSP implies a precision you do not have. Monte Carlo over the inputs you genuinely cannot pin down — here yield, glucose price and capital, each as a triangular distribution over its plausible range — gives a distribution instead: P10 $1.96, median $2.21, P90 $2.49, and a 1% probability of coming in below the incumbent's $1.80. That last number is the honest answer to "is this process viable at 50 g/L", and it is more useful than either optimism or a flat no.

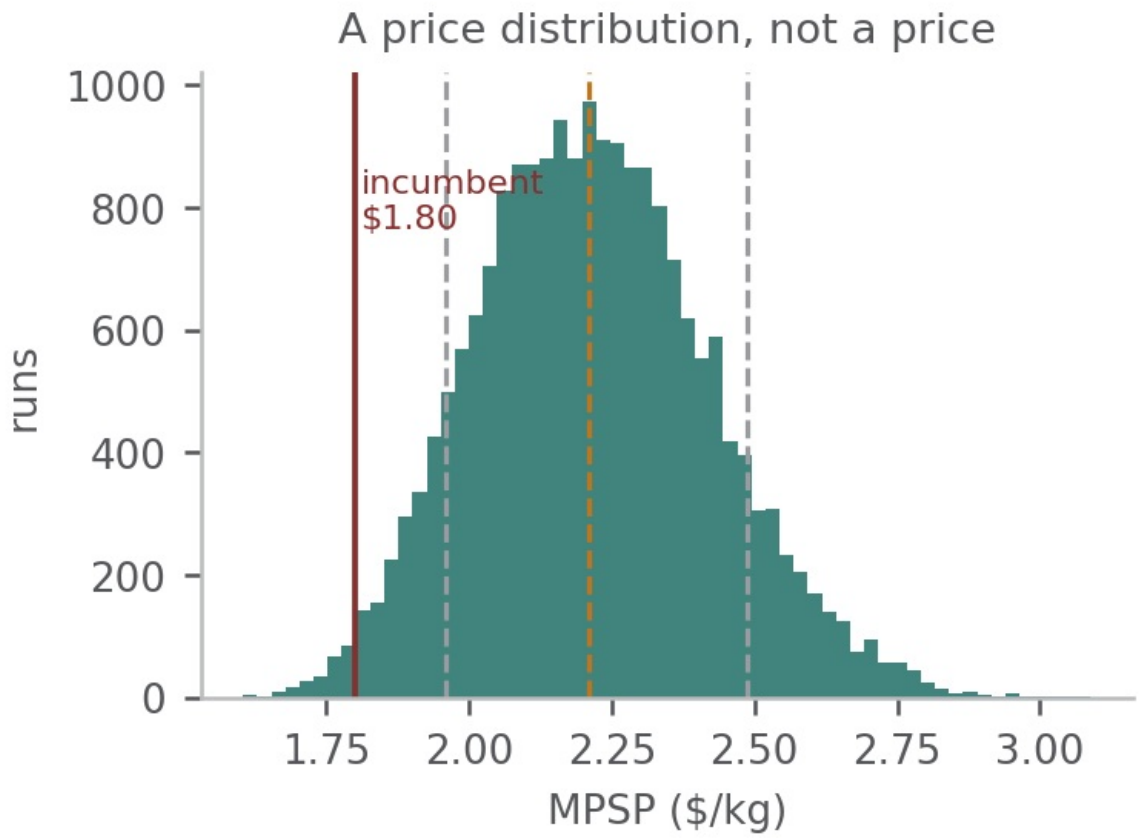


Figure 49.2 Twenty thousand Monte Carlo runs at 50 g/L over yield, glucose price and capital, with P10, median and P90 marked. The distribution sits almost entirely above the assumed incumbent price: the process does not fail because of an unlucky assumption, it fails structurally at this titer. Computed for this module.

Then the constructive step, which distinguishes a useful analysis from a discouraging one: **name the assumption carrying the conclusion and say what experiment retires it**. Here, yield — because unlike the glucose price it is ours to change, and because the model says each 0.05 g/g is worth about $0.13/kg. The experiment that retires it is a carbon balance on the current strain showing where the other 60% of the carbon goes, which is a week's work and changes what the next eighteen months are spent on.

Life-cycle assessment, and why the boundary decides the answer

Three definitions carry the whole subject. The **functional unit** is what you are comparing — one kilogram of product at a stated purity, not "a process". The **system boundary** says what is counted: cradle-to-gate means from the field or well to your factory gate, excluding use and disposal. **Allocation** decides how burdens are split when a process makes more than one saleable thing, and it is where honest people disagree: by mass, by energy content, by economic value, or by system expansion, and the choice can move the answer by a factor of two.

A first pass from your own mass and energy balances takes ten minutes. With glucose at an assumed 0.6 kg CO_2e/kg, a yield of 0.40 g/g contributes 1.50 kg CO_2e per kilogram of product; with 2.5 kWh/kg of electricity for agitation, compression and separation, at 0.40 kg CO_2e/kWh, energy contributes 1.00. Total 2.50 kg CO_2e/kg against a fossil incumbent assumed at 3.0 — a 17% advantage.

Now notice how fragile that is. A grid at 0.6 kg/kWh instead of 0.4 erases it entirely. Counting the nitrous oxide from fertiliser on the corn, or land-use change, raises the feedstock term. Giving the spent biomass credit as animal feed lowers it. The boundary choice, not the biology, decides whether the headline says 17% better or 20% worse — which is why reviewers go to the boundary first, and why your own LCA paragraph should state what it excludes before anyone asks.

Reading a published TEA critically

Four questions, in order. **What yield and titer does it assume, and are they achieved or projected?** Many published analyses model a target strain rather than an existing one, which is legitimate and must be stated. **What does the separation train cost, and was it designed or assumed?** Downstream is often half the capital and is frequently a single line item. **What feedstock price, and from what year?** Sugar prices swing by a factor of two, and

an analysis built on a 2015 price is a different analysis today. **What is the discount rate and plant life?** Moving from 10% over 20 years to 15% over 10 changes the capital charge per kilogram by about 70% and can be buried in a methods sentence.

Capital recovery	$CRF = i(1+i)^n/((1+i)^n - 1)$; 0.1175 at 10%, 20 y
MPSP	feedstock + utilities + fixed + CRF·capex/annual output
Feedstock term	substrate price ÷ yield; moved by yield only
Capital term	moved by titer and rate, through vessels and broth volume
Equipment scaling	$C = C_0(S/S_0)^{0.6}$, then an installation factor; Lang is a first pass
LCA first pass	feedstock CO_2e/yield + energy CO_2e per kWh × kWh/kg

Your strain is at 35 g/L and 0.28 g/g. Your supervisor wants a year of fermentation optimisation to reach 70 g/L. What does the model say, and what do you propose?

Run both. At 0.28 g/g the feedstock term alone is \$0.40/0.28 = \$1.43/kg, so even with capital and fixed costs driven to zero by an infinitely good titer the process cannot clear \$1.80 once utilities and labour are added — doubling the titer takes MPSP from roughly \$3.00 to \$2.30 and still loses money. Raising the yield to 0.40 g/g at the current titer of 35 g/L takes the feedstock term to \$1.00 and MPSP to roughly \$2.40; doing both gets to about \$1.75 and clears. So the answer is that neither alone is sufficient and yield is the one that must happen.

What I would propose: the carbon balance first, because at 0.28 g/g — probably half of theoretical — the carbon is going somewhere identifiable, and by-product measurement costs a week against a year of fermentation optimisation. Then the titer campaign, with the TEA rerun at each milestone so that the project has a stated pass mark rather than a direction. And the sentence that makes this a collaboration rather than an argument: the model says a year of titer work raises the price we can tolerate by \$0.70/kg, which is worth doing *after* the yield work and not instead of it.

Three things people get wrong

- ‘TEA is for the business school.’ It is the only calculation that ranks titer against yield against rate, which is the central strain-engineering decision.
- ‘The MPSP is \$2.16.’ It is a distribution; report P10, median and P90 and the probability of clearing the market price.
- ‘Our process is greener because it is biological.’ The boundary and the grid decide that, and a first-pass number from your own balances will tell you in ten minutes.

Self-check — answer on paper first

1. Explain why doubling the titer reduces cost but doubling the yield reduces it differently, in terms of the four cost lines.

Titer reduces the number of vessels needed for a given output and the volume of broth the separation train must handle, so it moves the capital charge and the fixed costs that scale with plant size; it leaves the feedstock term untouched, because the substrate consumed per kilogram of product is unchanged. Yield moves only the feedstock term, which in a commodity process is usually the largest single line. They are complementary rather than alternatives, and which to pursue first depends on which line currently dominates — which the breakdown tells you directly.

2. A published TEA reports \$1.20/kg for a product you know is made at 15 g/L. What do you look for?

Almost certainly a projected rather than achieved performance, so first the assumed titer and yield in the methods rather than the abstract. Then the separation train, since at 15 g/L the broth volume per kilogram is 67 litres and the downstream capital should be large; if it is a single assumed line item, that is where the \$1.20 comes from. Then the discount rate and plant life, and whether co-products have been credited by an allocation choice that moves most of the burden elsewhere. None of these makes the paper wrong; all of them are needed to know what it claims.

3. Your LCA gives 2.5 kg CO_2e/kg against a fossil incumbent at 3.0. Why is that not yet a claim you can publish?

Because the comparison rests on boundary and allocation choices that have not been justified: whether land-use change and fertiliser emissions are in the feedstock figure, what grid intensity the electricity is costed at, whether spent biomass receives a co-product credit, and whether the two routes deliver the same functional unit at the same purity. A 17% advantage is well inside the range those choices move the answer, so the publishable version states the boundary, shows the result under the two or three defensible allocation choices, and says which of them the conclusion survives.

What to open a book for. No book, but real software: install BioSTEAM, run a Bioindustrial-Park configuration unchanged,

then build your own. Cortes-Peña et al., ACS Sustain. Chem. Eng. (2020) is the paper to cite and to read for what the platform does with uncertainty. Peters, Timmerhaus and West, or any process-economics text, covers capital estimation properly if you want the engineering rather than the API.

Analytical method validation: trusting your own numbers

Every conclusion in your thesis is a number from an instrument, multiplied by a calibration, corrected by a recovery, in a matrix that is not water. This week is about the multiplication.

Calibration, and the fit that lies to you at the bottom of the range

Build the curve from a dilution series spanning the concentrations your samples actually occupy, plot the residuals, and ignore R². Here is why, computed rather than asserted. Take a genuinely linear detector — true slope 12,500 area units per g/L, true intercept 150 — with noise that is 4% proportional plus a small constant term, which is what a chromatographic detector really does. Standards from 0.05 to 50 g/L. Fit it two ways.

	Slope	Intercept	Back-calculated error at 0.05 / 0.1 / 0.25 g/L
True values	12,500	150	—
Unweighted least squares	11,767	3,431	−551% / −260% / −98%
Weighted (1/σ²)	12,382	229	−11% / +7% / +6%

Both fits have R² above 0.999. The unweighted fit is dominated by the 25 and 50 g/L standards, because their absolute residuals are hundreds of times larger than those at the bottom, and it pays for its fit up there with an intercept of 3,431 — which, when you back-calculate a sample at 0.05 g/L, returns a negative concentration. A curve that reports negative concentrations for real samples is not a rare pathology; it is the normal consequence of unweighted regression over two orders of magnitude.

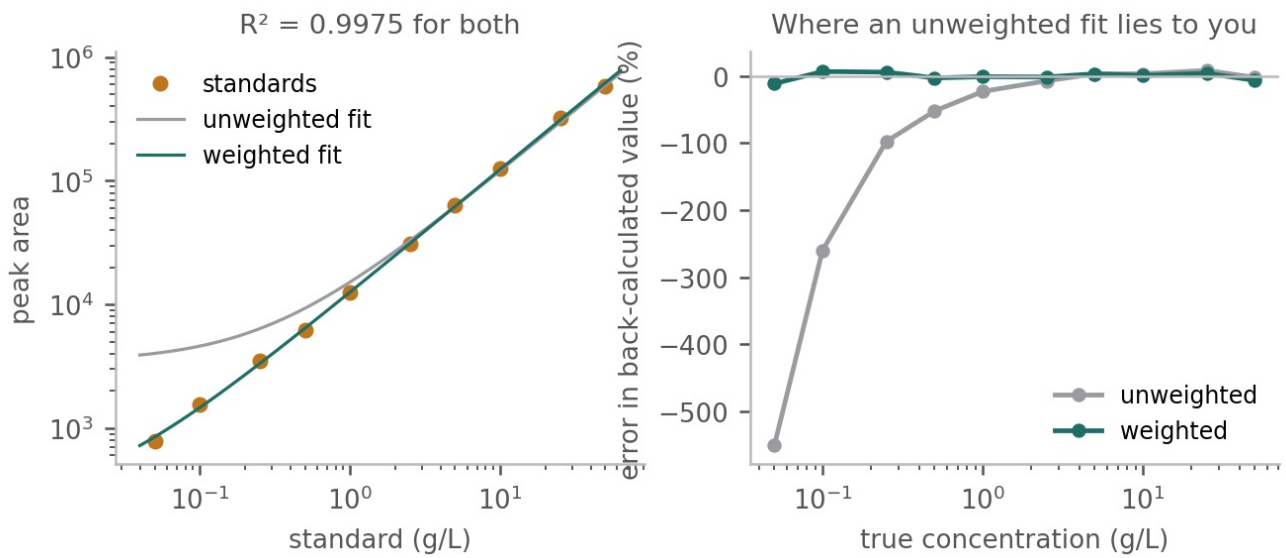


Figure 50.1 Left: the same ten standards fitted with and without weighting; both have R² above 0.999 and the lines are indistinguishable on a log plot. Right: the error in the back-calculated concentration of each standard. The unweighted fit is unusable below about 1 g/L. Synthetic data with 4% proportional noise, computed for this module.

Three rules follow. **Weight by 1/σ²** when the error is proportional — in practice 1/x or 1/x² weighting, which every chromatography data system offers and almost nobody switches on. **Do not force through zero** unless you have evidence the intercept is zero; forcing it hides exactly the baseline offset and carryover that a non-zero intercept is telling you about. And **state the working range**, with a rule for what happens outside it: samples above the top standard get diluted and re-injected, samples below the quantification limit get reported as below it, not as a number.

Limits of detection and quantification, and the standard abuse

With σ the standard deviation of the blank response, or of the residuals near zero, and m the calibration slope, the conventional definitions are

```
LOD = 3.3·σ/m        LOQ = 10·σ/m
```

In the example above, σ ≈ 60 area units and m = 12,382, so LOD = 0.016 g/L and LOQ = 0.048 g/L. What each permits you to claim is different and the difference is the point. Above the LOD you may say the analyte is present. Above the LOQ you may say how much. Between them you have evidence of presence and no measurement — and the abuse, which you will see in print constantly, is a table of values at a third of the LOQ carried through to a yield calculation with three significant figures and an error bar derived from triplicate injections of the same unmeasurable sample.

The honest handling of a trace result is to report it as < LOQ, to say what the LOQ is, and — if the trace matters to the conclusion — to concentrate the sample, change the method, or state that the experiment cannot answer the question. A negative result at a stated LOQ is a measurement; a small number below the LOQ is not.

Accuracy, precision, and four words that mean four experiments

Repeatability is the spread of replicate injections of the same sample, on the same instrument, in one session: it measures the instrument. **Intermediate precision** is the spread across days, operators, columns and calibration curves: it measures your laboratory, and it is typically two to five times worse than repeatability. Quoting repeatability where intermediate precision is meant is how a method comes to look better than the numbers it produces over a six-month project.

Accuracy is closeness to the true value and cannot be assessed by replication at all, only by recovery of a known addition. **Recovery** is the fraction of a spike you get back, and the single most important instruction of this week is that it must be measured in fermentation broth and not in water. Broth contains salts, protein, antifoam, residual sugar, pigments and cell debris, all of which suppress or enhance ionisation, bind analyte, co-elute, or foul the column. A method that recovers 99% in water and 62% in broth is a method that reports 62% of your titer, and the only way to find out is to spike the broth.

Where matrix effects are strong — which for mass-spectrometric quantification of intracellular metabolites is always — an internal standard becomes necessary rather than good practice, and the right one is an isotope-labelled version of the analyte: it experiences the same suppression, the same extraction losses and the same instrument drift, and the ratio cancels them. A structurally different internal standard cancels injection volume and little else.

Before the instrument: where the sample is actually lost

For an intracellular measurement, most of the error happens before anything is injected. Metabolite pools turn over in seconds — chapter 13's argument for the pseudo-steady state, read backwards — so the sample must be quenched faster than that, usually into cold methanol, and every known quenching method has an artefact: cold methanol leaks small metabolites out of the cells, fast filtration takes time, and whole-broth extraction measures intracellular and extracellular pools together. There is no method without a loss; there is only a method whose loss you have characterised.

The rule that makes this tractable is the one worth memorising: **a 30% extraction loss applied evenly is survivable; a loss that varies between strains invalidates the comparison**. A systematic 30% under-recovery shifts every pool by the same factor, so ratios, fold changes and flux interpretations survive. But if your producing strain has a different membrane composition, a different cell size or a different lysis behaviour from the control — which engineered strains often do — then the recovery differs between the two and the comparison is measuring extraction, not biology. Hence: measure recovery separately in each strain being compared, and say so in the methods.

Carryover, drift and the sample queue

Two instrument behaviours corrupt long runs, and both are revealed by queue design rather than by inspection. **Carryover** is analyte from the previous injection appearing in the current one; detect it by placing a blank immediately after your highest standard and quantifying what appears in it, expressed as a percentage of the high standard. **Drift** is a slow change in response across a run; detect it by placing a quality-control sample of known concentration every ten injections and plotting its measured value against position in the queue.

Then the design that protects the conclusion: randomise the order of samples across treatments, so that a drift cannot align with your experimental groups. This is week 17's randomisation argument applied to the instrument instead of the plate, and it is the reason a queue that runs all the controls first and all the engineered strains afterward can manufacture a difference out of a drifting detector. Blocking works too — alternate control and test — and is easier to defend.

The method section, as a checklist

A methods section a stranger can evaluate contains: instrument and column with dimensions and particle size; mobile phase and gradient, or oven programme; detector and wavelength or ion transitions; standards, their source and purity,

and how the stock was prepared; internal standard and its concentration; calibration range, number of levels, replicates, weighting, and whether the intercept was forced; LOD and LOQ with how they were determined; recovery in matrix, by strain if strains are compared; repeatability and intermediate precision with the experiments that produced them; the queue structure including blanks, QCs and randomisation; and sample handling — quenching, extraction, storage temperature and time to analysis. It is a long list and most of it is one line each.

Weighted calibration	`weight 1/σ² (in practice 1/x or 1/x²) when error is proportional`
Limits	`LOD = 3.3σ/m, LOQ = 10σ/m; 0.016 and 0.048 g/L in the example`
Precision	`repeatability = one session; intermediate = days, operators, columns`
Accuracy	`only from recovery of a known spike, in broth and not in water`
Extraction loss	`even loss survivable; strain-dependent loss invalidates the comparison`
Queue	`blank after the top standard (carryover), QC every 10 (drift), randomise order`

A collaborator reports that strain B makes 0.31 g/L against strain A's 0.09, a 3.4-fold improvement, from triplicate injections with an R^2 of 0.9997 calibration. What do you ask, and what is your guess?

Ask four things. Whether the calibration was weighted and what the LOQ is — if the LOQ is near 0.1 g/L, strain A's value is below it and the fold change is a ratio involving an unmeasurable number. Whether the triplicates are injections or independent cultures; three injections of one culture is n = 1 in the week 17 sense. Whether recovery was measured in broth for both strains, since a strain-dependent difference in extraction would produce exactly this result. And whether the two strains were run in the same queue and in what order.

My guess, in order of likelihood: the low value sits below the quantification limit, so the true statement is "strain A was below the LOQ of 0.1 g/L and strain B measured 0.31 g/L", which is still a real and publishable improvement but is a different claim with a different error structure. Second most likely is an unweighted calibration making the low end unreliable. The improvement is probably real; the number 3.4 probably is not, and the fix — rerun both with a weighted low-range calibration, independent cultures and a matrix spike — costs two days.

Three things people get wrong

- ‘R^2 = 0.999, so the calibration is good.’ R^2 is insensitive to exactly the low-end bias that ruins trace measurements. Plot residuals, and weight the fit.
- ‘Below the LOQ but we reported it anyway.’ Below the LOQ you have presence, not a value. Report it as such.
- ‘Recovery was 98%.’ In water. Broth is not water, and the number you need is per strain if strains are being compared.

Self-check — answer on paper first

1. Why does an unweighted calibration over two orders of magnitude systematically mis-measure the low end, and what does the symptom look like?

Because least squares minimises absolute residuals, and with proportional error the absolute residuals at the top of the range are orders of magnitude larger than at the bottom, so the top standards determine the line. The fit buys accuracy there at the cost of an inflated intercept, and the symptoms are a positive intercept far larger than the blank, back-calculated standards at the bottom that are badly wrong or negative, and real low-concentration samples reporting negative concentrations — all while R^2 remains excellent.

2. Design the experiment that establishes recovery for an intracellular metabolite in two strains being compared.

Spike a known amount of the analyte — ideally the isotope-labelled version, so it is distinguishable from what the cells contain — into the cell suspension of each strain immediately before quenching, then carry both through the full quench, extraction and analysis. Recovery is the measured spike over the added spike, computed separately for each strain, with replicates. Spiking after extraction instead measures only matrix suppression at the instrument, which is also worth knowing: the ratio of the two spikes separates extraction loss from ionisation suppression, and reporting both is what a careful methods section does.

3. Your QC samples drift downward by 12% across a 90-sample queue. What does that do to your results, and what should have been done?

If the samples were ordered by treatment, it manufactures a difference of up to 12% between groups that is purely instrumental, and if the real effect is of that size the conclusion is unsafe. The corrections available after the fact are to normalise against the QC trend or against an internal standard, both of which assume the drift affects samples and QCs

identically. What should have been done is randomisation or blocking of the run order, plus the internal standard — which would have cancelled most of the drift automatically and turned a salvage operation into a non-event.

What to open a book for. No textbook; the sources are a chromatographic method-validation tutorial and a review of quenching and extraction for microbial metabolomics. Read one published methods section in your own product area alongside them and list what it omitted — that exercise teaches more than either source, and the list becomes your own template.

Omics, screening, and turning data into model constraints

An omics dataset is not evidence about flux. It becomes useful at the moment you can say which parameter or bound in a model it sets — and for three of the four layers, the honest answer is none directly.

The four layers, and what each cannot tell you

- **Transcriptomics** measures mRNA abundance. It is cheap, genome-wide and fast, and it measures the layer furthest from flux: mRNA predicts protein only loosely, protein predicts activity only loosely, and activity predicts flux only in combination with substrate concentrations. What it is genuinely good for is detecting a regulatory response — a stress programme firing, a catabolite repression you did not expect, a promoter silenced in stationary phase.
- **Proteomics** measures protein abundance, and is the first layer that constrains anything quantitatively, because an enzyme's abundance times its turnover number is an upper bound on the flux it can carry. Absolute quantification is harder than relative, and relative is what most datasets are.
- **Metabolomics** measures pool concentrations. It cannot give flux — a pool can be large and static or small and fast — but it gives two things nothing else does: the mass-action ratios that decide thermodynamic feasibility and direction (chapter 14), and the accumulating intermediate that localises a bottleneck.
- **Fluxomics**, meaning ^{13}C metabolic flux analysis, measures flux itself, including the split ratios and reversibilities that every stoichiometric model has to assume. It is the only layer that is evidence about what happened rather than about what the cell was equipped to do, and it is the most laborious.

For a strain that grows well and makes nothing, measure metabolites first. Growth being normal rules out a gross burden or energy problem, so the question is where in the installed pathway the carbon stops — and an accumulating intermediate answers it directly and cheaply, while a transcript dataset would tell you that your genes are expressed, which you already believe.

Analytics, and the throughput that matters

HPLC with UV or refractive index detection is the workhorse for extracellular titers, sugars and organic acids: robust, quantitative, and slow at a few minutes to half an hour per sample. GC-MS needs volatility or derivatisation and is the standard for fatty acids, terpenes, alcohols and derivatised amino acids. LC-MS covers polar, charged and thermally fragile compounds and is what intracellular metabolomics is done on, at the cost of matrix effects severe enough to make chapter 50's isotope-labelled internal standards mandatory. The sample-preparation step that most often ruins metabolomics is the quench, for the reasons in the last chapter.

Then screening throughput, which is the week-11 test bottleneck quantified:

Format	Throughput	What it measures
Shake flask with HPLC	tens per week	the product, properly, with yields
Microtitre plate with a colorimetric or fluorescent proxy	10^3–10^4 per day	a correlate of the product
Biosensor-coupled FACS	10^7 per hour	a transcription-factor response to an intracellular metabolite
Growth coupling or selection	10^9 and above	survival, which you have arranged to require production

Five orders of magnitude separate the top and bottom rows, and each step down the table trades fidelity for throughput: the thing measured gets further from the thing you want. That trade is the reason the library-coverage arithmetic of chapter 17 matters — a 10^6 library can only be searched by the bottom two rows — and the reason every selection campaign needs the confirmation rate reported, because cells optimise what is selected and will find ways to light up a biosensor without making product.

From a dataset to a constraint: the arithmetic

Here is the calculation that turns proteomics into modelling, and it is one line. An enzyme's maximum flux is its abundance times its turnover number:

```
v ≤ k_cat · [E]
```

Put real units on it. Cell protein is about 0.5 g per gram of dry cell weight. An enzyme at 1% of that protein, with a molecular weight of 50 kDa, is present at 5 mg/gDCW = 1×10^{-7} mol/gDCW. With $k_{cat} = 10\ s^{-1}$, the ceiling is 10^{-6} mol·s^{-1} per gram of cells, which is **3.6 mmol·gDCW^{-1}·h^{-1}**. Invert it and you have the more useful form: carrying 10 mmol·$gDCW^{-1}$·h^{-1} through an enzyme with that k_{cat} requires 2.8% of the cell's entire protein, for one step.

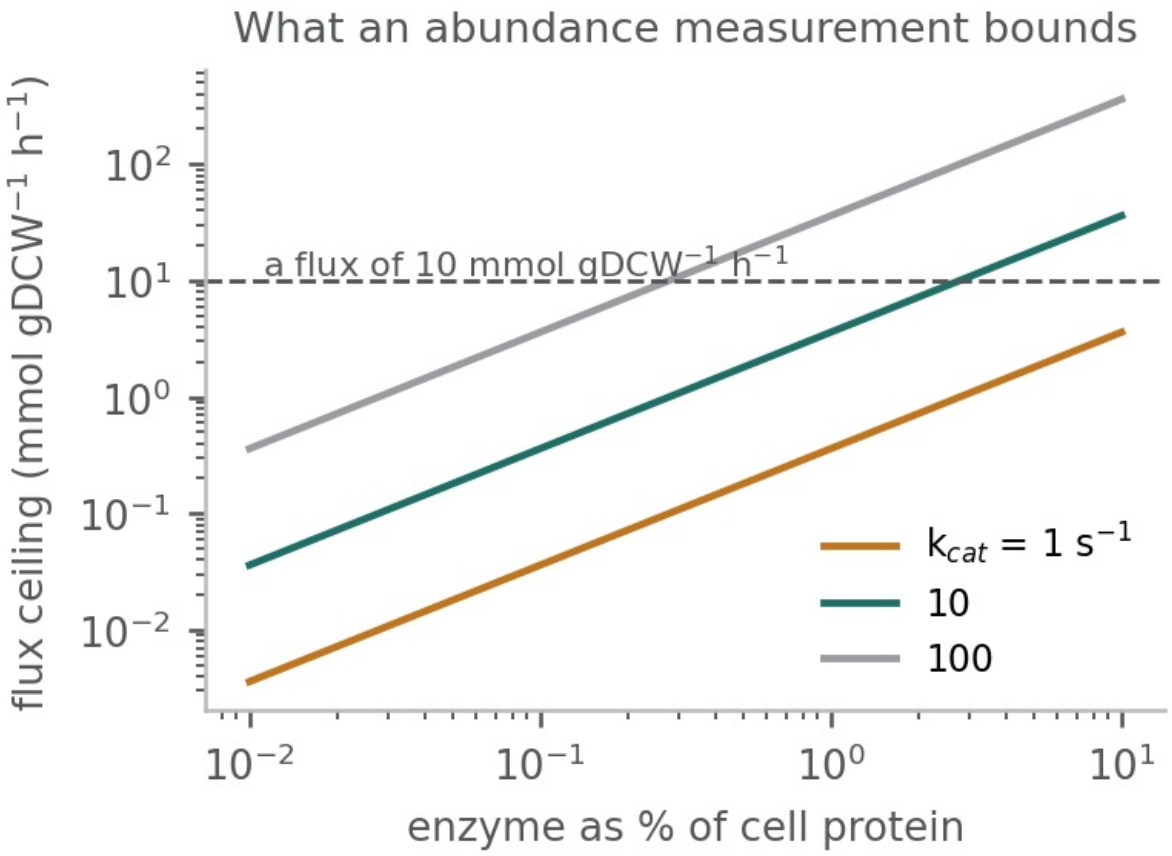


Figure 51.1 The flux a single enzyme can carry, against its share of cell protein, for three turnover numbers, assuming 0.5 g protein per gDCW and a 50 kDa enzyme. The dashed line is a flux of 10 mmol·$gDCW^{-1}$·h^{-1}, roughly a tenth of glycolytic flux in a fast-growing cell. Arithmetic, computed for this module; k_{cat} and molecular weight are the assumptions.

Two things follow immediately. First, a low-k_{cat} enzyme is not merely slow, it is expensive: at $k_{cat} = 1\ s^{-1}$ the same flux needs 28% of the proteome, which is impossible, so the pathway cannot run at that rate however hard you express it. That is a design rejection you can make from a single literature k_{cat}, before any cloning. Second, summing the requirement across all enzymes gives the proteome budget — $\Sigma\ v_i \cdot MW_i / k_{cat,i} \le$ total protein — which is what enzyme-constrained models such as GECKO add to flux balance analysis. That single extra constraint is what makes a constraint-based model predict overflow metabolism and the growth laws of week 9 instead of needing them imposed by hand, because fermentation is cheaper in protein per ATP than respiration.

Layer by layer: what each can and cannot set

Layer	Can set	Cannot set
Transcriptomics	Which reactions to switch off in a condition-specific model; a qualitative on/off prior	Any flux bound quantitatively — the mRNA-to-flux chain has two lossy steps in it
Proteomics	An upper bound $v \le k_{cat}[E]$ per reaction, and the total proteome budget	The actual flux, which may be far below the bound; and k_{cat} is not measured here
Metabolomics	Mass-action ratios, hence $\Delta G'$ and direction bounds; saturation state against K_M	Flux, and the distinction between a static pool and a fast one
Fluxomics	The fluxes themselves, with split ratios and reversibilities	Much beyond central metabolism, where labelling information runs out

The honest caveats on the proteomics route are worth stating because they decide how much to trust the resulting model. The k_{cat} values come from in vitro assays, often on the enzyme from a different organism, at a temperature and

pH unlike the cytoplasm, and they can be wrong by an order of magnitude. Absolute protein quantification requires labelled standards or careful intensity-based methods. And an upper bound is not a prediction: an enzyme present at 1% of protein and carrying 1% of its ceiling is perfectly normal, so a model constrained this way tells you what is impossible rather than what is happening.

Which brings the phase 3 checkpoint's best question: the difference between a flux that FBA calls feasible and one that ^{13}C-MFA calls actual. FBA returns a flux vector consistent with stoichiometry, the bounds you imposed and an assumed objective — a hypothesis about what could happen. ^{13}C-MFA infers fluxes from measured labelling patterns — evidence about what did happen, including the split ratios and reversibilities a stoichiometric model must assume. A design justified only by FBA should be validated by labelling before it is believed, and the sentence to be able to say is that the two methods answer different questions and the second is the one that can falsify a strain hypothesis.

Enzyme ceiling	`v ≤ k_cat·[E]; 1% of protein at k_cat 10 s⁻¹ → 3.6 mmol gDCW⁻¹ h⁻¹`
Protein cost of a flux	`fraction of proteome = v·MW/(k_cat · 0.5 g protein per gDCW)`
Proteome budget	`Σ v_i·MW_i/k_cat,i ≤ total protein — the GECKO constraint`
Screening ladder	`flask 10¹/wk · plate 10³–10⁴/d · FACS 10⁷/h · selection 10⁹`
FBA against ^{13}C-MFA	`feasible under an assumed objective, against inferred from labelling`

Your strain's titer is a third of what your FBA predicts. Proteomics shows the pathway enzymes are abundant and transcriptomics shows the genes are strongly expressed. What do you conclude, and what do you measure next?

That the limitation is not expression, which is what those two datasets establish and the only thing they establish. Four candidates remain, and they are distinguishable. The enzymes may be abundant and inactive — misfolded, missing a cofactor, or mis-assembled — which an in vitro activity assay on the lysate settles in a day. The pathway may be thermodynamically or kinetically stalled at a particular step, which metabolomics settles by showing an accumulating intermediate and a mass-action ratio near equilibrium. The carbon may not be arriving, which a ^{13}C labelling experiment or simply a carbon balance with by-products settles. Or the FBA prediction may be wrong, which is likely if it assumed growth maximisation for a burdened engineered strain.

Order: lysate activity first, because it is a day and it bisects the space — abundant and active sends you to supply and thermodynamics, abundant and inactive sends you to expression conditions and cofactors. Then metabolite pools across the pathway. Then labelling, which is weeks of work and should be spent only once the cheaper layers have narrowed the question. And note what the enzyme-constraint arithmetic adds: if the abundant enzymes are carrying far less than $k_{cat}\cdot[E]$ allows, the limitation is definitely not the enzymes, which is a useful thing to be able to state quantitatively.

Three things people get wrong

- ‘The transcriptomics shows the pathway is upregulated, so flux increased.’ Two lossy steps sit between mRNA and flux. It shows regulation responded.
- ‘Metabolomics gives us the flux.’ It gives pool sizes. A large pool can be static and a small one fast; flux needs labelling.
- ‘An enzyme-constrained model predicts the flux.’ It bounds it. A model built from upper bounds tells you what is impossible.

Self-check — answer on paper first

1. Compute the fraction of cell protein needed to carry 5 mmol·gDCW^{-1}·h^{-1} through a 60 kDa enzyme with k_{cat} = 2 s^{-1}, and say what you would do with the answer.

The flux is $5 \times 10^{-3}/3600 = 1.39 \times 10^{-6}$ mol·s^{-1}·gDCW^{-1}, so the enzyme needed is that divided by k_{cat}, 6.9×10^{-7} mol/gDCW, which at 60 kDa is 42 mg/gDCW — about 8% of the cell's 0.5 g of protein, for one step. That is a serious but not impossible burden, and what I would do with it is treat the enzyme as the first candidate for protein engineering or replacement by a faster homologue, because every doubling of k_{cat} halves a proteome cost that is already competing with growth.

2. Why does adding a proteome budget to flux balance analysis make it predict overflow metabolism without being told to?

Because respiration requires far more protein per ATP produced than fermentation does — the respiratory chain, the TCA cycle and their enzymes are a large investment — so once total protein is a constrained resource, the optimal solution above a threshold uptake rate is to make part of the ATP by the protein-cheap fermentative route and excrete the product. Plain FBA, which has no protein constraint, sees no reason to waste carbon and predicts full respiration. The

overflow is therefore a consequence of the constraint rather than an observation added to the model, which is exactly the proteome-allocation account of week 9.

3. You can run either one ^{13}C-MFA experiment or twenty plate-based screens this quarter. How do you decide?

By what the current bottleneck is. If you have many untested designs and no idea which works, the screens buy parallelism and the labelling buys a detailed picture of a strain you may abandon. If you have one strain whose behaviour you cannot explain — product low with no accumulating intermediate and no by-product — the labelling buys the only evidence that will settle where the carbon goes, and twenty more screens will produce twenty more unexplained strains. The general rule from chapter 11 is that parallelism beats depth when the prior is poor, and depth beats parallelism when a single well-characterised failure would change the design.

What to open a book for. Ko, Lee et al., Chem. Soc. Rev. 49:4615 (2020) is the review to read for the omics layers and how a systems-metabolic-engineering group actually uses them. For enzyme-constrained models, read the GECKO methodology paper rather than a textbook; for ^{13}C-MFA, week 43's sources. No book covers the translation from dataset to constraint, which is why the table in this chapter exists.

Host defences and pooled screens: getting DNA into a wild strain, and reading 50,000 of them at once

A wild industrial strain has spent its evolutionary life defending itself against incoming DNA, and it is good at it. The arithmetic of how good takes five minutes and tells you whether your construct can survive the journey.

Why this week is optional, and when it stops being optional

The tracker marks this optional because a project in *E. coli*, *S. cerevisiae* or another laboratory host never meets the problem: those strains have had their restriction systems removed or characterised decades ago, and the cloning kits assume it. The week becomes mandatory the moment you work with a wild isolate, an industrial production strain, a non-model host from week 25, or anything a collaborator describes as "the strain our group uses" without being able to name its genotype. In those cases it is the first week to do, not the last, because the alternative is three months of transformations that return nothing and are blamed on competence.

Restriction–modification, in four lines

- **Type I.** A single complex that both methylates and cleaves, recognising a bipartite asymmetric site and cutting at a distance from it, often thousands of base pairs away. Hard to predict, and the cleavage position means site removal from the construct is not a reliable defence.
- **Type II.** Separate enzymes for cleavage and methylation, recognising a short usually palindromic sequence and cutting within or beside it. These are the familiar cloning enzymes and the easiest case: known motif, predictable sites, removable by codon-level sequence changes.
- **Type III.** A combined complex that requires two inversely oriented recognition sites to cut, so a construct carrying only one site survives — which is why orientation matters when you count sites.
- **Type IV.** The inverse logic: it cleaves DNA that *is* methylated, in specific patterns. So DNA prepared in a standard *E. coli* cloning strain, carrying Dam and Dcm methylation, can be destroyed precisely because it was methylated — and the fix is a methylation-free cloning host, which is the opposite of the Type I and II fix.

REBASE lists the systems found in sequenced genomes along with their recognition motifs, so the first action for a new host is to look it up and write down the motifs. If the genome is sequenced but the systems are only predicted, treat the prediction as a hypothesis: methyltransferase genes are easy to recognise and their specificities are not.

How often a motif occurs, and why a 6-base cutter is fatal

This is the calculation of the week. For a random sequence of equal base composition, a specific k-base motif occurs with probability $(1/4)^k$ at each position, and a double-stranded plasmid of length L offers about $2(L - k + 1)$ positions once both strands are counted. For a 6 kb plasmid:

Motif length	Expected sites in 6 kb	P(no sites at all)
4 bp	46.9	4×10^{-21}
5 bp	11.7	8×10^{-6}
6 bp	2.93	0.054
7 bp	0.73	0.48
8 bp	0.18	0.83

Read the third column, which is the probability that an unmodified plasmid gets past that one system untouched, taking the counts as Poisson. A single 6-base system gives your construct about a 5% chance; a 4-base system gives it none at all. And strains carry several systems — three independent 6-base specificities take the survival probability to $0.054^3 \approx$ **0.0002**, which is why the number of distinct systems matters more than the identity of any one of them. A strain with four or five systems destroys essentially everything foreign, and a transformation efficiency of a few colonies per microgram against 10^8 in *E. coli* is not poor technique, it is the defence working.

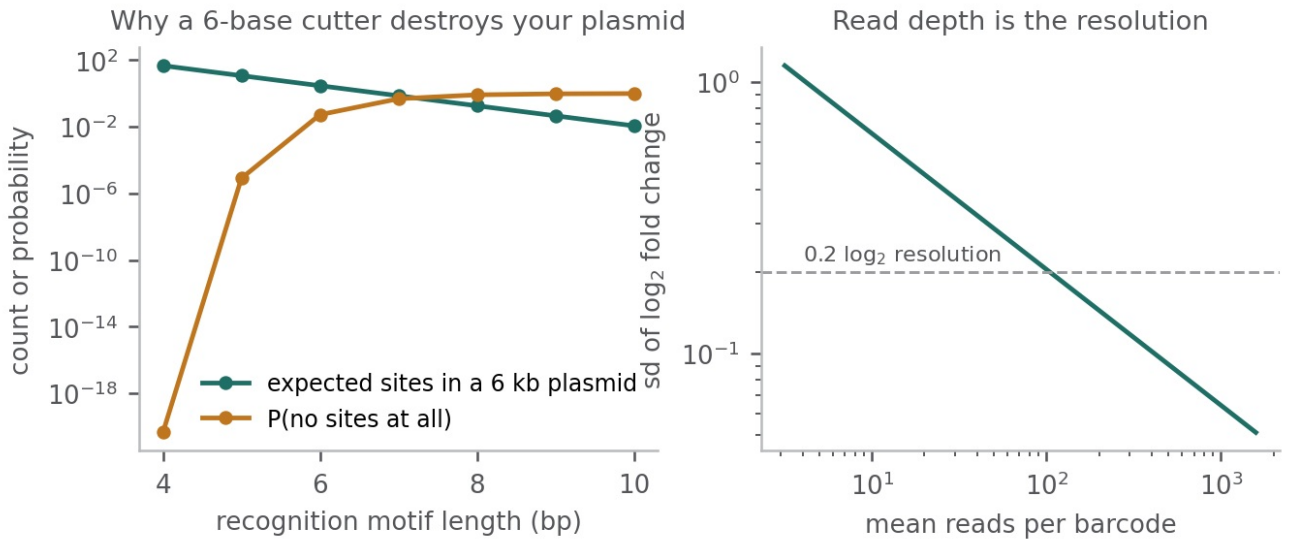


Figure 52.1 Left: expected sites and the probability of a site-free 6 kb plasmid against recognition-motif length, from the Poisson approximation to random sequence. Right: the standard deviation of a $\log_2$ fold change estimated from barcode counts at a given read depth, from counting statistics alone. Both computed for this module; the left panel assumes equal base composition, which real plasmids and real GC-skewed genomes violate, so treat it as the order of magnitude.

Getting past the barrier, and confirming that you did

Four routes, with different costs.

- **Methylate the plasmid in advance.** Express the host's own methyltransferases in your cloning strain, so the DNA arrives already marked as self. This is the general-purpose answer, it works against Types I to III, and it needs the methyltransferase genes cloned, which is a project of a few weeks and is reusable forever afterward.
- **Remove the sites.** Synonymous codon changes and non-coding edits to eliminate Type II motifs. Cheap when sites are few and the systems are Type II with known motifs; useless against Type I, whose cleavage is remote, and impossible when the motif count is in the dozens.
- **Change the cloning host.** Against Type IV, prepare the DNA in a *dam*$^{-}$ *dcm*$^{-}$ strain so there is no methylation to attack. One day of work, and the first thing to try if the host has a Type IV system.
- **Change the delivery.** Conjugation delivers single-stranded DNA, which some restriction systems handle poorly, and electroporation of large amounts of DNA can saturate the defences. Conjugation is the standard route into *Streptomyces* for exactly this reason, as week 10 noted.

The confirmation is a controlled comparison, and it is worth insisting on: transform the same plasmid, prepared methylated and unmethylated, side by side, and compare transformants per microgram. A tenfold to thousandfold difference is the defence being bypassed; no difference means the barrier was not your problem and you should look at competence, the origin of replication, the selection marker or the promoter instead. Without that control, a successful transformation after four changes tells you nothing about which change mattered.

Pooled screens: what a barcode buys

The second half of the week is a different subject with the same motivation — scale. A pooled screen constructs a library, attaches a sequencable barcode to each member, grows the whole pool under a condition, and reads the barcode abundances before and after. The fitness of each member is the log fold change in its count.

The structural insight, which is what the random barcoded transposon work made practical, is that barcoding **decouples library construction from fitness measurement**. You map barcodes to insertion sites once, expensively, by sequencing junctions; thereafter every experiment is a cheap amplicon sequencing of the barcodes alone. So the marginal cost of asking a new question of the same library is a PCR and a sequencing lane, and a library that cost months can answer dozens of questions.

What it cannot do is also structural. A pooled screen measures the average behaviour of each genotype in a mixed population, so it cannot see anything that depends on the clone being alone: product titer in its own vessel, morphology, anything cross-fed between neighbours — a mutant that cannot make a metabolite will be rescued by its neighbours and score as fit. It cannot report non-fitness phenotypes at all unless they are coupled to growth or to a sortable signal, and it cannot distinguish a genotype that grows well from one that grows slowly and lyses little. Those questions need isolated clones, which is why the last day of the week is about converting a pooled hit into a validated one.

Counting barcodes, and the read depth the question needs

Counts are Poisson, so precision follows from depth with no biology in it. For a barcode observed at mean depth d in each of two conditions, the standard deviation of the estimated $\log_2$ fold change is approximately $\sqrt{(2/d)}/\ln 2$:

Mean reads per barcode	sd of $\log_2$ fold change	Reads needed for a 50,000-member library
10	0.65	0.5 million
50	0.29	2.5 million
100	0.20	5 million
500	0.09	25 million

So detecting a twofold fitness effect — one $\log_2$ unit — reliably needs only tens of reads per barcode, while detecting a 15% effect, 0.2 $\log_2$ units, needs hundreds, and that is before biological variation between replicate cultures, which is usually larger than the counting noise and is the reason replicate pools matter more than extra depth. The practical sequence: decide the effect size that would change your decision, read off the depth, multiply by the library size, and compare with what a lane delivers. It is the same pre-experiment arithmetic as chapter 17's power calculation, wearing different clothes.

Analysis then follows week 33's RNA-seq machinery exactly, because the data have the same shape: normalise for sequencing depth and composition, model counts rather than ratios, shrink fold changes for low-count barcodes, and control the false discovery rate across thousands of tests rather than the per-test error. A volcano plot of fitness against significance is the standard output, and the barcodes with large effects and low counts — the top-left of the plot — are where the false positives live.

From a pooled hit to a phenotype you believe

A hit from a pooled screen is a hypothesis about one genotype, measured in competition, with a mutation whose identity comes from a barcode map. Validating it has three steps and they are not optional. **Reconstruct cleanly**: rebuild the mutation in the parent background, because the library member may carry passengers. **Measure it alone**: the phenotype in isolation is the one you will report, and it frequently differs from the competitive one — a mutant can be fit in a pool because of what its neighbours provide. **Measure it in competition with the parent**: this reproduces the screen's own measurement and tells you whether the pooled result was real.

How many replicates the follow-up needs comes straight from chapter 17: for an effect of about one pooled standard deviation, n = 3 has roughly a third of the power you want, so plan for six to eight independent cultures if the decision matters. And decide in advance what would make you drop the hit — for instance, no effect in isolation and no effect in a clean competition, which together say the barcode was mis-mapped or the member carried a passenger mutation. Writing that pass mark down before the follow-up is what stops a hit surviving on sunk cost.

Expected sites	`2(L − k + 1)·(1/4)^k; 2.93 for a 6-mer in 6 kb`
Site-free plasmid	`P ≈ e^(−expected); 0.054 for one 6-base system, 0.0002 for three`
Bypass	`pre-methylate (I–III), site removal (II only), dam⁻dcm⁻ host (IV), conjugation`
Confirmation	`transformants per µg, methylated against unmethylated, same prep`
Barcode precision	`sd(log₂FC) ≈ √(2/depth)/ln 2; 0.20 at 100 reads per barcode`
Sequencing budget	`library size × depth; 50,000 × 100 = 5 million reads per sample`

A collaborator's industrial *Bacillus* strain takes your plasmid at 30 colonies per microgram against 10^8 in *E. coli*. Plan the next two weeks.

Week one is arithmetic and lookup, not bench work. Pull the strain's genome from REBASE or, if it is not listed, annotate the restriction–modification genes and note their predicted motifs and types. Count the occurrences of each motif in the plasmid, on both strands and with orientation recorded for any Type III system. Compare the counts with the table above: if there are two Type II systems with a few sites each, the predicted survival probability is percent-level, which matches an efficiency of 30 per microgram and means the barrier is almost certainly the explanation rather than competence. If the strain has a Type IV system, note that your standard cloning prep is methylated and is therefore the problem.

Week two is the bypass and its control. The cheapest first test is a *dam*$^-$ *dcm*$^-$ prep, which costs a day and resolves the Type IV case outright. In parallel, order synthesis of the host's methyltransferase genes for a methylating helper plasmid in the cloning strain — the general solution, reusable for every subsequent construct in this host. Then run the comparison that proves it: the same plasmid, prepared both ways, transformed side by side, efficiency per microgram reported for each. If efficiency rises by orders of magnitude the barrier is beaten; if it does not, the problem was never restriction, and the next suspects are the replication origin, the marker and the promoter — all of which are cheaper to test once restriction has been excluded by a control rather than by hope.

Three things people get wrong

- ‘Low transformation efficiency means poor competent cells.’ In a wild strain it usually means the defences are working, and the arithmetic will tell you before the bench does.
- ‘We removed the restriction sites.’ Which works for Type II and not for Type I, whose cleavage happens far from the recognition site, nor for Type IV, which attacks methylation rather than sequence.
- ‘The pooled screen says this gene is important.’ In competition, in a pool, averaged over a genotype. Reconstruct it and measure it alone before believing it.

Self-check — answer on paper first

1. Compute the expected number of sites for a 6-base motif in a 10 kb plasmid and the probability it carries none, and say what you would conclude from a strain with two such systems.

About $2 \times 10{,}000/4{,}096 = 4.9$ expected sites, so the probability of none is $e^{-4.9} \approx 0.007$. With two independent systems of that kind the plasmid's chance of arriving intact is roughly 5×10^{-5}, which means essentially every molecule is cut and any transformants you obtain arrived in rare unmethylated-and-uncut molecules or by repair. The conclusion is that the construct cannot be delivered unmodified and that pre-methylation or a different delivery route is required, not more electroporation.

2. Why does barcoding decouple library construction from fitness measurement, and what does that change about experimental design?

Because the expensive step — determining which barcode corresponds to which mutation — is done once by junction sequencing, after which the identity of every member is known from a short barcode amplicon. Each subsequent experiment therefore costs a PCR and a sequencing lane rather than a library build. What it changes is that the marginal cost of a new condition collapses, so the design should plan many conditions against one library rather than one condition against many libraries, and the limiting resource becomes the conditions you can think of and the replicate cultures you can run, not the cloning.

3. You want to resolve a 15% fitness difference in a 20,000-member library. What depth and how many reads, and what will actually limit you?

Fifteen per cent is about 0.2 $\log_2$ units, so from counting statistics you need the standard deviation below roughly that, which takes about 100 reads per barcode — 2 million reads per sample for 20,000 members, which is trivial on any modern instrument. What will actually limit you is biological variation between replicate pools, which at this effect size exceeds the counting noise, so the budget should go into independent replicate cultures rather than deeper sequencing of the same ones. That is the same conclusion as chapter 17's biological-against-technical replicate argument.

What to open a book for. REBASE is the reference and it is a database, not a book. Wetmore et al., mBio (2015) is the RB-TnSeq paper and is worth reading in full for the design logic rather than the protocol. For the restriction barriers, a review of restriction–modification in non-model bacteria; Snyder & Champness, from week 10, remains the best background on how these systems were worked out.

Closing notes to part 4

The four weeks in one page

Week 49. Minimum product selling price is the number that ranks titer against yield against rate. In the model here, 20 / 50 / 100 g/L give $3.22 / $2.16 / $1.80 per kilogram against a $1.80 incumbent, with feedstock unmoved by titer and capital unmoved by yield. Report a distribution, name the assumption carrying the conclusion, and say what experiment retires it. For the life-cycle number, the boundary decides the answer, so state it before anyone asks.

Week 50. An unweighted calibration over two decades is unusable at the bottom — in the worked example, −551% error at 0.05 g/L with R^2 above 0.999. Above the detection limit you may say present; above the quantification limit you may say how much. Recovery is measured in broth, per strain when strains are compared, and an even extraction loss is survivable while a strain-dependent one invalidates everything.

Week 51. Omics becomes useful when it sets a bound. An enzyme at 1% of cell protein with $k_{cat} = 10\ s^{-1}$ can carry 3.6 $mmol \cdot gDCW^{-1} \cdot h^{-1}$, and carrying 10 would take 2.8% of the whole proteome — which is how a single k_{cat} rejects a design before cloning, and how enzyme-constrained models predict overflow instead of assuming it. Transcriptomics sets nothing quantitative; metabolomics sets thermodynamics; only labelling measures flux.

Week 52. A single 6-base restriction system gives an unmodified 6 kb plasmid a 5% chance of arriving intact, and three such systems reduce it to 0.02%. The bypass depends on the system type and the confirmation is a methylated-against-unmethylated efficiency comparison. In pooled screens, barcoding makes new conditions nearly free; counting statistics set the read depth, and biological replicates set everything else.

On the numbers in this part

The techno-economic model is listed in full in chapter 49 and every parameter in it is an assumption — vessel size, batch time, installed capital, utilities, discount rate — so the output is a structure rather than a forecast, and the tornado and Monte Carlo panels exist to show which parts of the structure carry weight. The calibration demonstration uses synthetic data generated with 4% proportional noise; the slopes and errors quoted are what the fits returned. The enzyme-constraint arithmetic assumes 0.5 g protein per gram of dry cell weight and a 50 kDa enzyme, and k_{cat} values in the real literature are frequently wrong by an order of magnitude, which is the dominant uncertainty in any enzyme-constrained model. The restriction-site counts assume random sequence with equal base composition, which real plasmids violate, so they are order-of-magnitude guides rather than site counts — count the actual motifs in your actual sequence, which takes one line of code.

What is still to come

Working forward in tracker order, the remaining weeks with nothing behind them are the product-class weeks 53 to 57 — pharmaceuticals and terpenoids, commodity chemicals and amino acids, biofuels and C1, the PHA case study, and CO_2, hydrogen and waste carbon as feedstocks — and then the host weeks: yeasts in 58, *C. glutamicum* and *B. subtilis* in 59, phototrophs in 60, and the panorama with cell-free systems in 61.

Those two blocks fit together, and that is how the next parts will be written. A product class is a set of constraints — the precursor it draws on, its cofactor demand, whether it is secreted, whether it is toxic, what price it must clear — and a host is a set of capabilities against exactly those constraints. So the product chapters will end in a constraint sheet for each class, and the host chapters will be a comparison against those sheets rather than four species descriptions: tools and editing efficiency, growth rate and medium cost, native precursor and cofactor supply, secretion and post-translational capability, tolerance, regulatory acceptance, and the specific arithmetic each host forces — compartment transport costs for the yeasts, which is chapter 48's dead-core calculation with membranes instead of gel, and light penetration for the phototrophs, which is the same calculation again with photons instead of oxygen.